

Programming Group Project 2024S1

Assignment 02

Version 1.0

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Chapter 01: Introduction

1.1 Overview

The rapid growth of digital technology has transformed the education sector, with a demand for low-cost, flexible, and affordable learning. Traditional classroom learning, while effective, is tainted by restricted access, rigid timetables, and cost, deterring many from pursuing further studies. To address these problems, we propose the development of Certify Learn, an e-learning system developed with existing web technologies such as HTML, CSS, Bootstrap, JavaScript, PHP, and MySQL. Certify Learn will give learners a hassle-free and trouble-free experience of learning with study materials, certification exams, and an easy-to-use interface. By combining free learning and low-cost certification, the site seeks to allow users to learn something new, certify knowledge they already have, and advance their careers easily and low cost.

The site is designed to be consumed by a broad range of consumers including students, careerists, and lifelong learners. It will have courses with varying levels of difficulty, from simple courses like “Introduction to PHP” and “Introduction to Java” to high-level ones. User interactive functions like reviews, feedback, and progress tracking will also be a part of Certify Learn to enhance the learning experience. Admin features for user administration, course administration, and quiz administration will also be strong for scalability and security purposes.

1.2 Background and Motivation

The traditional school system, although effective in many ways, will not work when applied to today's students. Geography, time, and cost can be a barrier to individuals getting a proper education. The COVID-19 pandemic helped further increase the need for online learning since physical classrooms went out of bounds, and online learning solutions made a beeline for the demand. Certify Learn is designed to address these challenges by providing one site where

customers can find free learning resources, sign up for certification exams, and engage in a global network of learners.

The idea of Certify Learn is the rising need for low-cost education. Most of the existing e-learning platforms, such as Coursera and Udemy, offer good content but request users to pay a lot for courses and certifications. This creates an entry barrier for those who cannot sustain such outlays. Certify Learn will beat this by offering free and low-cost learning materials and affordable certification tests, thus making education available to all regardless of finances. The site will also place high emphasis on interactivity through ratings, comments, and progress indicators to ensure that one gets to have an interactive as well as personalized learning experience.

1.3 Aim and Objectives

1.3.1 Aim

The ultimate aim of Certify Learn is to provide an easy, affordable, and rational e-learning portal that integrates the ancient art of traditional learning and the modern science of Internet learning. The site will also offer free study material, certification exams, and an easy-to-use interface to make the learning process easy.

1.3.2 Objectives

1. Free Learning Content: Provide top-notch free or non-expensive learning content, like Introduction to PHP, Introduction to Java, etc. The content will all be designed keeping in mind the intention to serve students of different levels of know-how, i.e., newcomers and experts.
2. Certification Exams: Offer certification exams that verify users' knowledge and ability. Users can register for exams, pay a nominal fee, and receive certificates upon successful completion. Exams will be designed to assess practical knowledge and enable users to acquire the skills to get hired.

3. **User Interaction:** Incorporate features like user feedback, reviews, and tracking of learning progress to make the website more interactive and user-friendly. The feature will allow users to post a comment on their experience, provide feedback, and track learning progress.
4. **Admin Control:** Make administrators' control of courses, quizzes, and users more streamlined. The admin interface will include adding and modifying courses, quiz administration, and tracking user activity.
5. **Scalability and Security:** Make the platform secure in order to protect payment transactions and user data, and scalable so that it can serve a growing number of users. Technologies and best practices of high caliber will be utilized to design the platform such that it becomes highly reliable and secure.

1.4 Summary

Certify Learn is designed to overcome the limitations of conventional learning by making e-learning accessible, cost-effective, and flexible. By offering free study guides and low-cost certification exams, the platform makes it easy to empower users to acquire new skills and validate their proficiency. Support for user interaction features and robust admin management options makes it easy to learn in an interactive and smooth manner. Current e-learning platforms will be explored in the next chapter to obtain best practices and define the value proposition of Certify Learn.

Chapter 02: System Analysis and Design

2.1 Introduction

under the system analysis and design, we had to collect system requirements and analyze the unique features using several diagrams such as use case diagrams, sequence diagrams, activity diagrams, class diagrams, component diagrams, and entity relationship diagrams.

2.2 Use Case Diagram

A use case diagram is a graphical representation that illustrates the possible interactions between users and a system. In the proposed e-learning platform, use case diagrams help identify how different users (such as users, administrators, and guests) interact with the system boundaries. These diagrams provide an overview of the key functions users can perform, such as browsing courses, making purchases, managing courses, and viewing educational content, ensuring that the system meets the needs of both users and administrators.

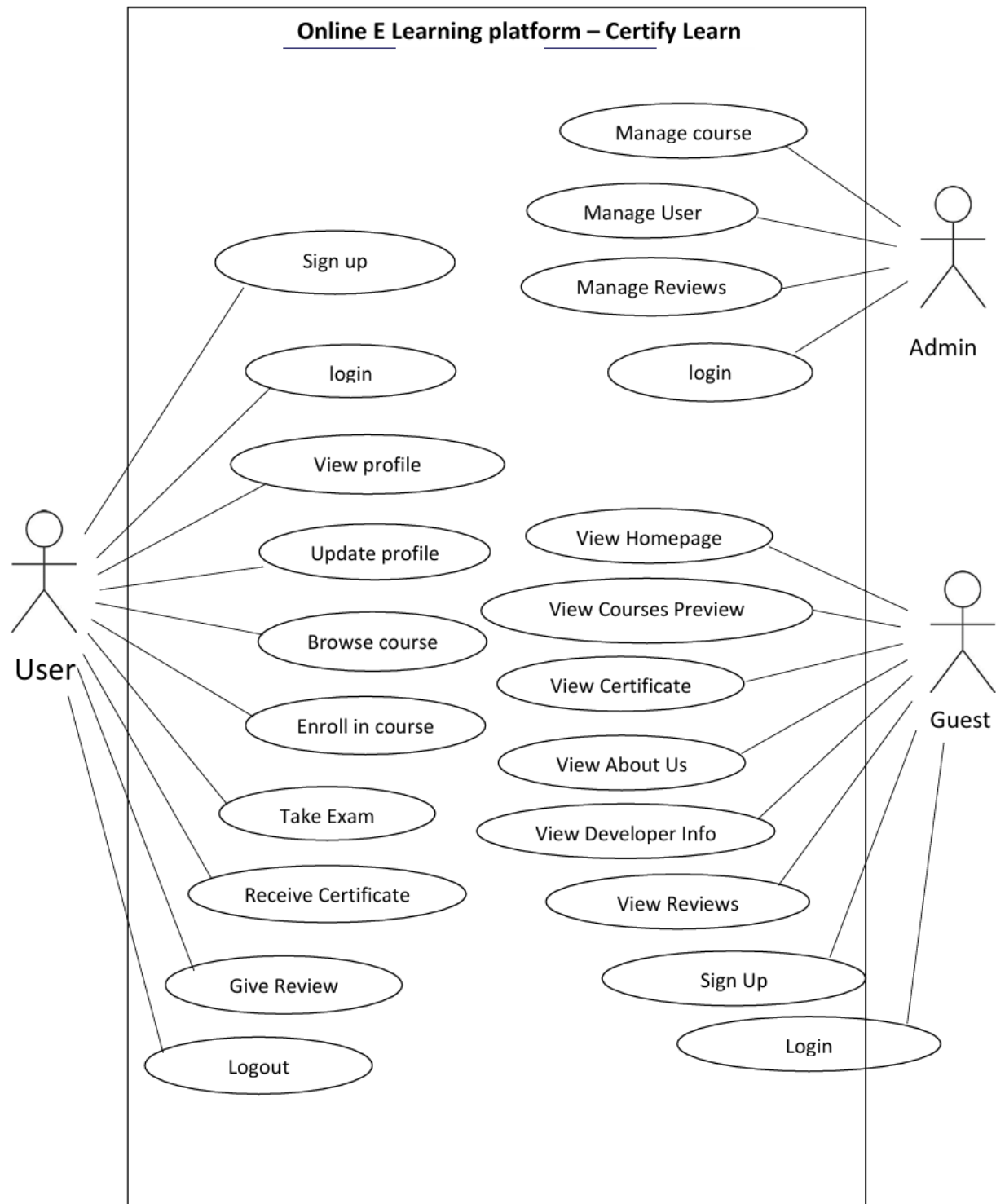


Figure 1: Use Case Diagram

2.3 Activity Diagram

An activity diagram visually represents a series of actions within a system. It serves as an advanced version of a flowchart, illustrating the flow of information and actions from one activity to the next. In the proposed e-learning platform, activity diagrams are used to map out the sequence of activities, such as browsing courses, making purchases, and processing payments. These diagrams help in identifying the flow of operations within the system, ensuring a seamless and user-friendly experience for users.

2.3.1 Activity Diagram – Login System

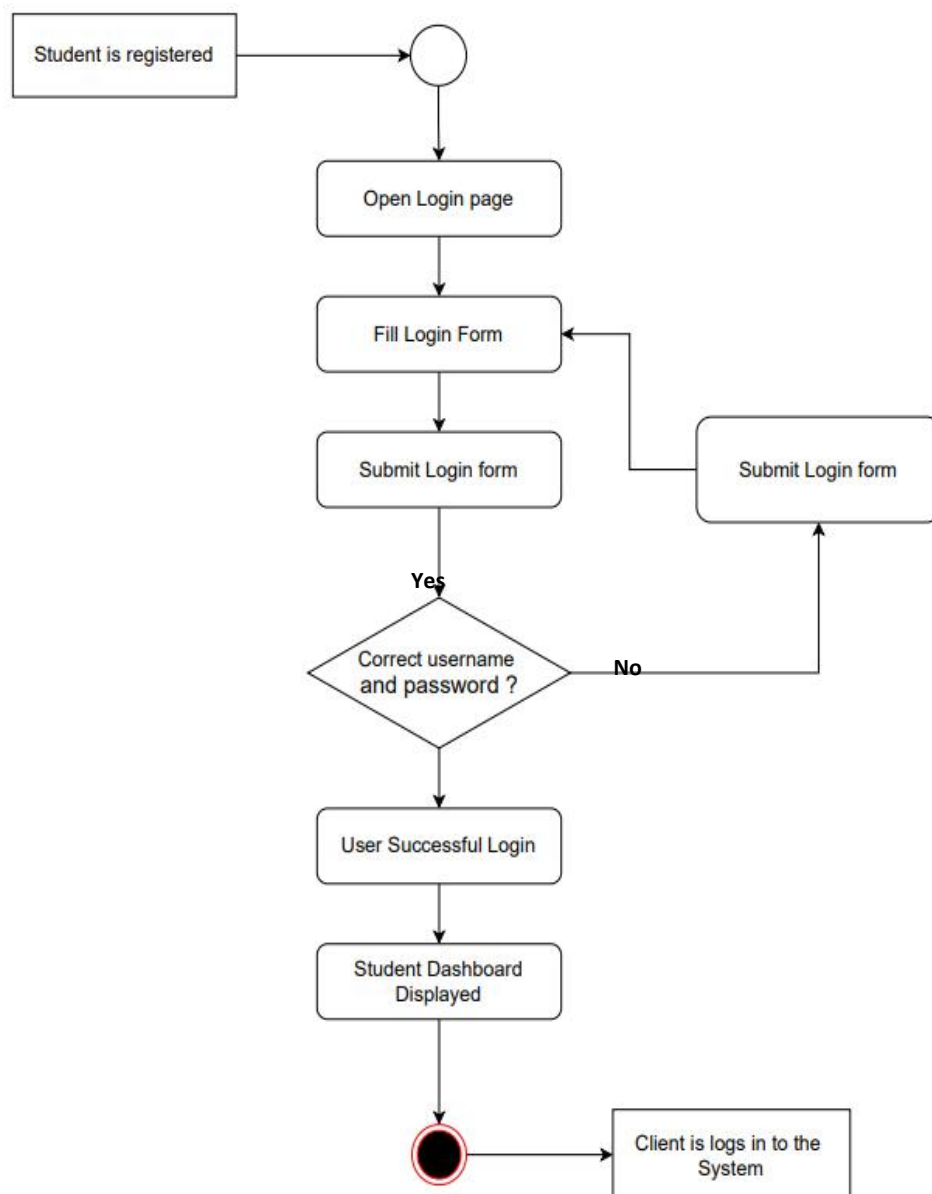


Figure 2: Activity Diagram for Login System

2.3.2 Activity Diagram – Admin Login

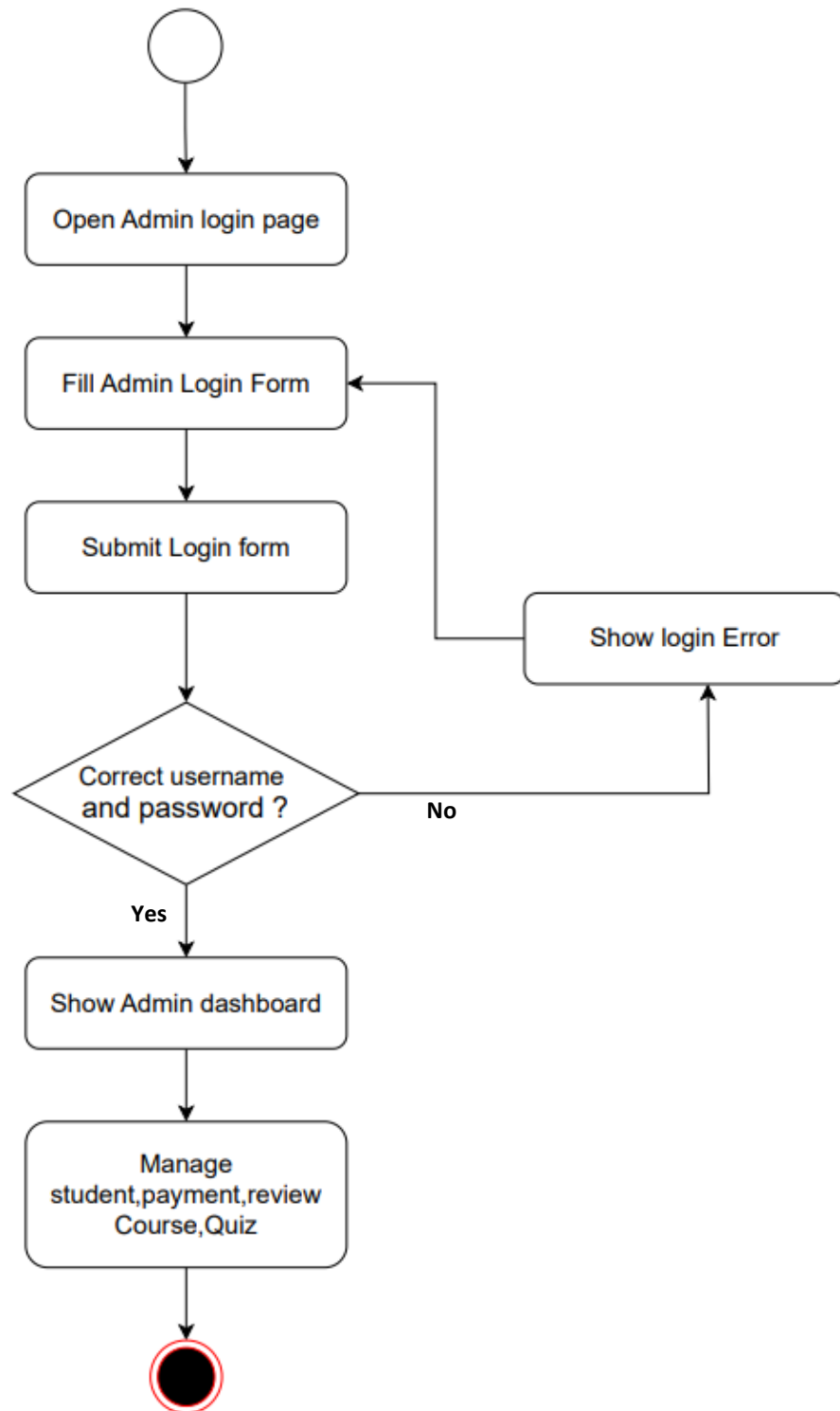


Figure 3: Activity Diagram for Admin Login

2.3.3 Activity Diagram – Course Management

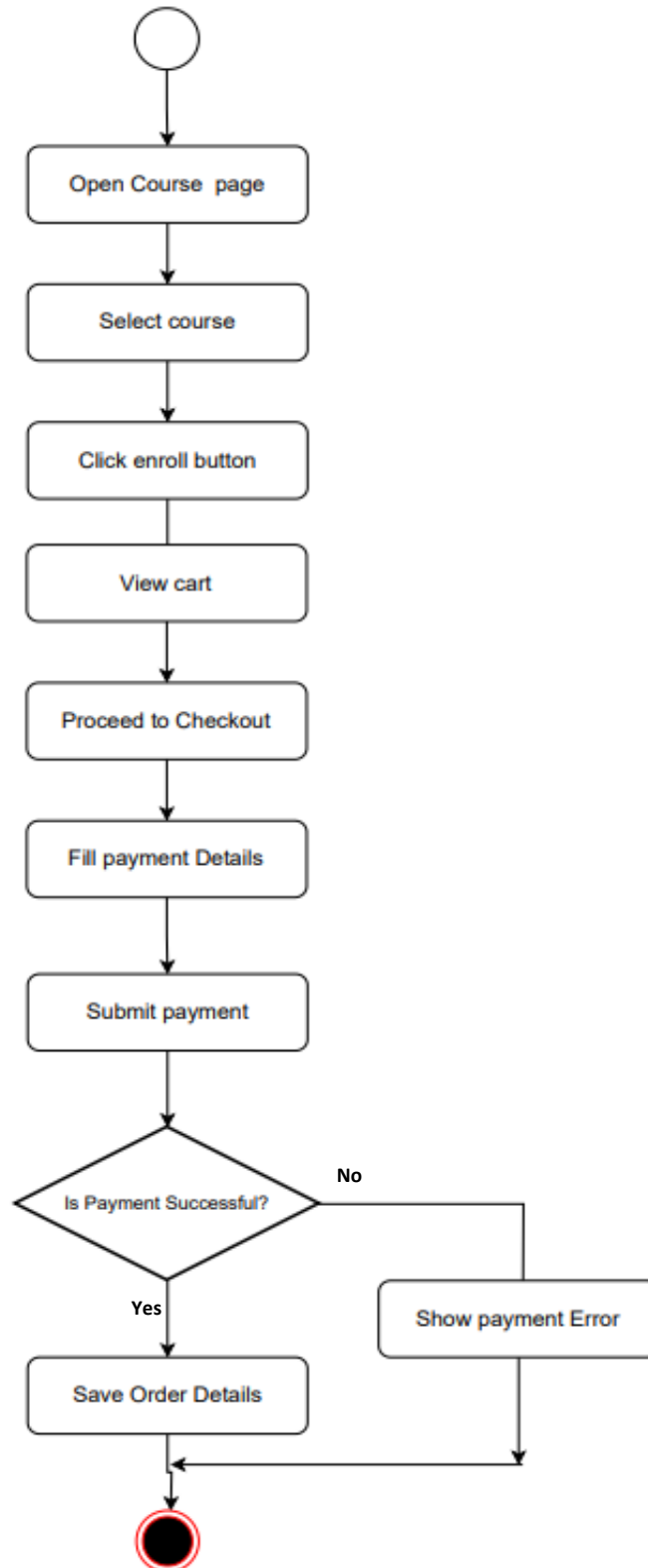


Figure 4: Activity Diagram for Course Management

2.3.4 Activity Diagram – Quiz Management

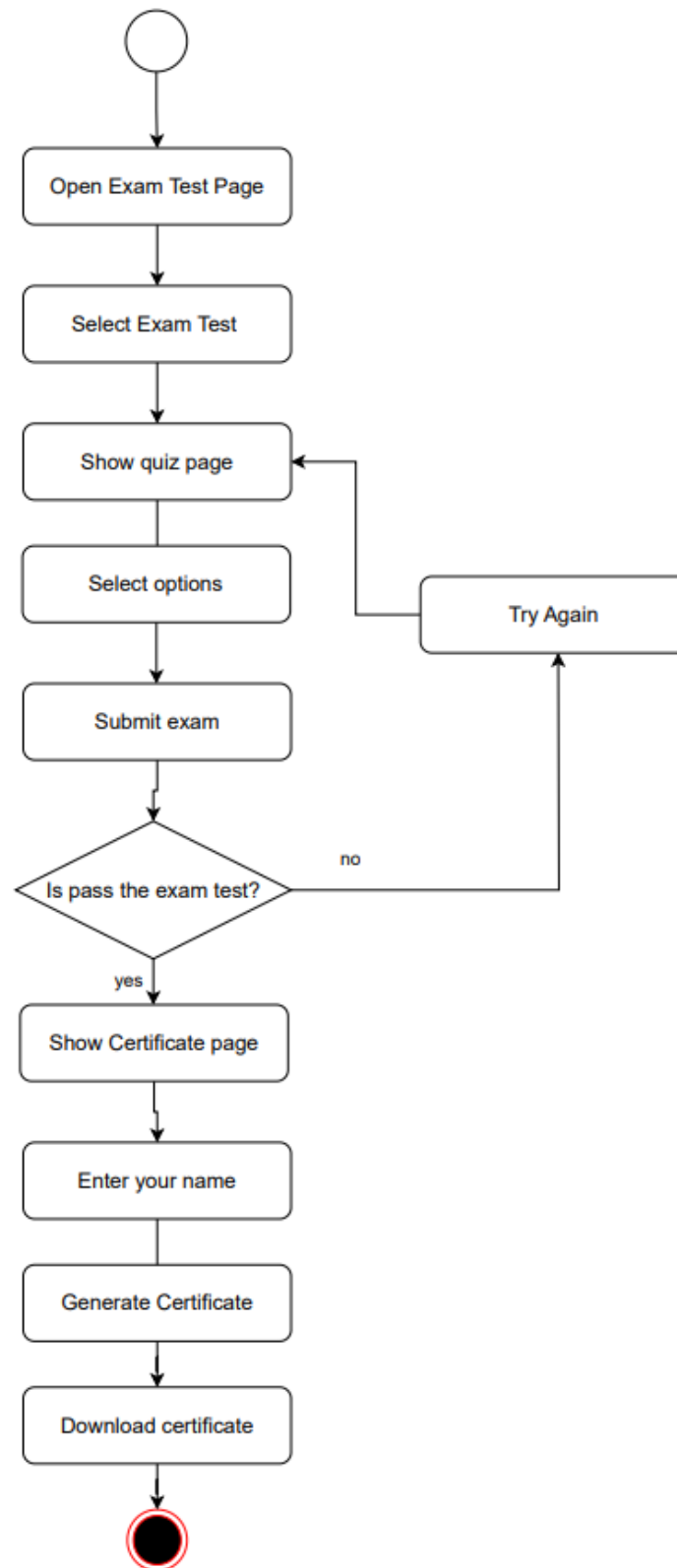


Figure 5: Activity Diagram for Quiz Management

2.4 Sequence Diagram

A sequence diagram is a diagram created in the Unified Modeling Language (UML) that illustrates the sequence of messages exchanged between objects during an interaction. Sequence diagrams excel at demonstrating how objects collaborate to achieve a particular outcome, though they may not define the behavior with precise detail. In the e-learning platform, sequence diagrams are used to capture the order of activities, such as browsing courses, adding courses to the cart, making payments, and receiving confirmation. These diagrams help visualize the step-by-step interaction between the system and its users.

2.4.1 Sequence Diagram for Admin

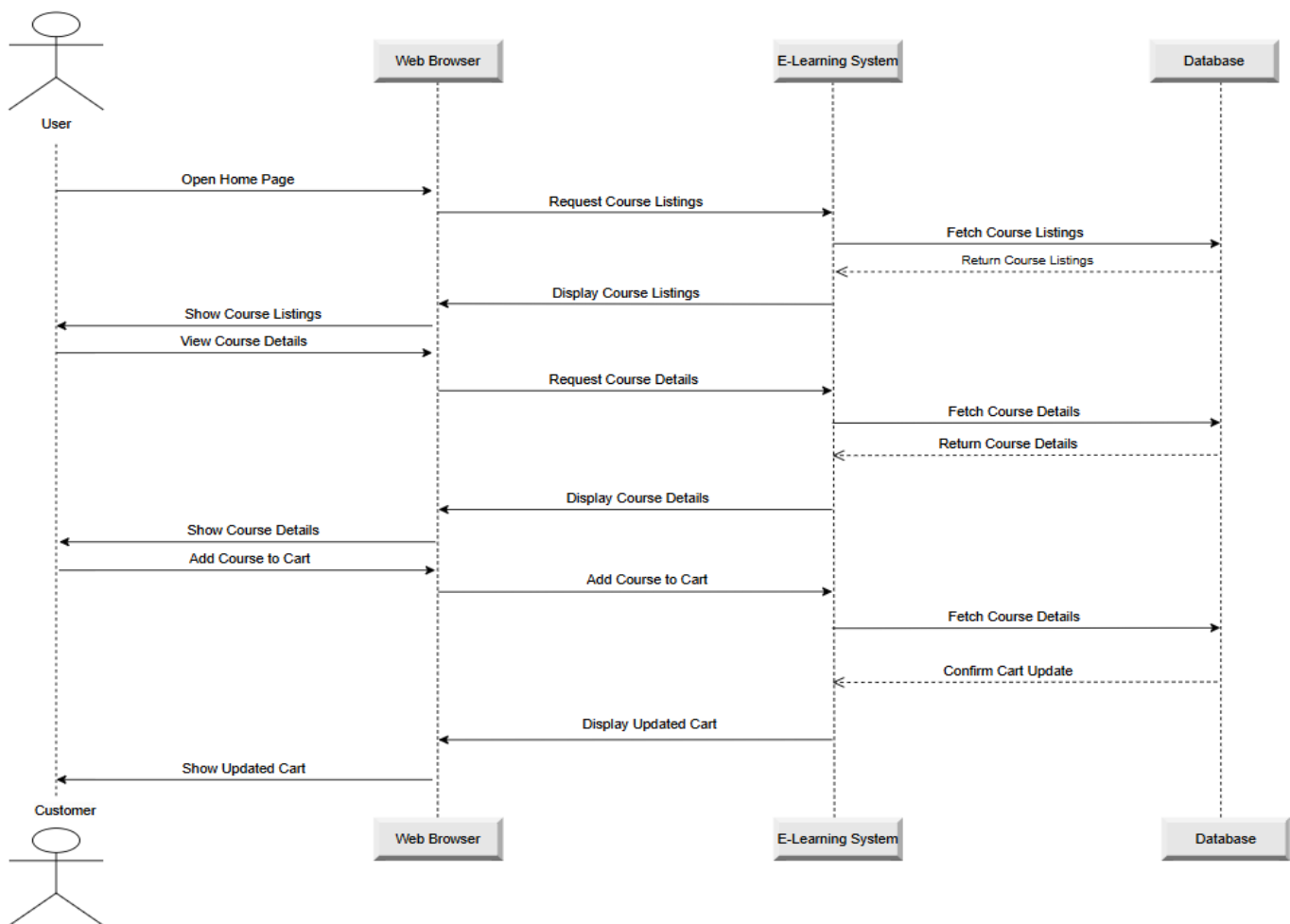


Figure 6: Sequence Diagram for Admin

2.4.2 Sequence Diagram for User Login and Registration

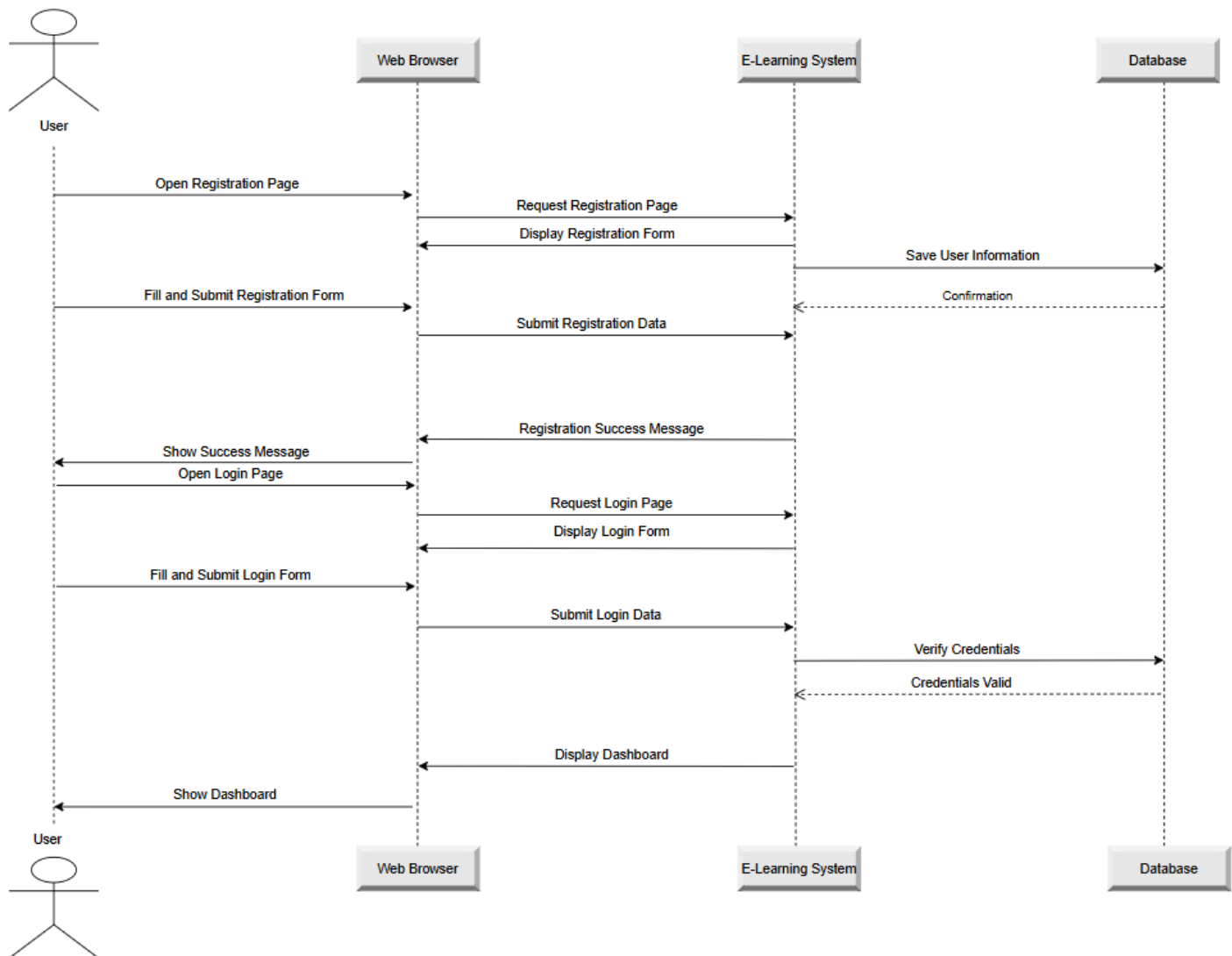


Figure 7: Sequence Diagram for User Login and Registration

2.4.3 Sequence Diagram for Course and Payment

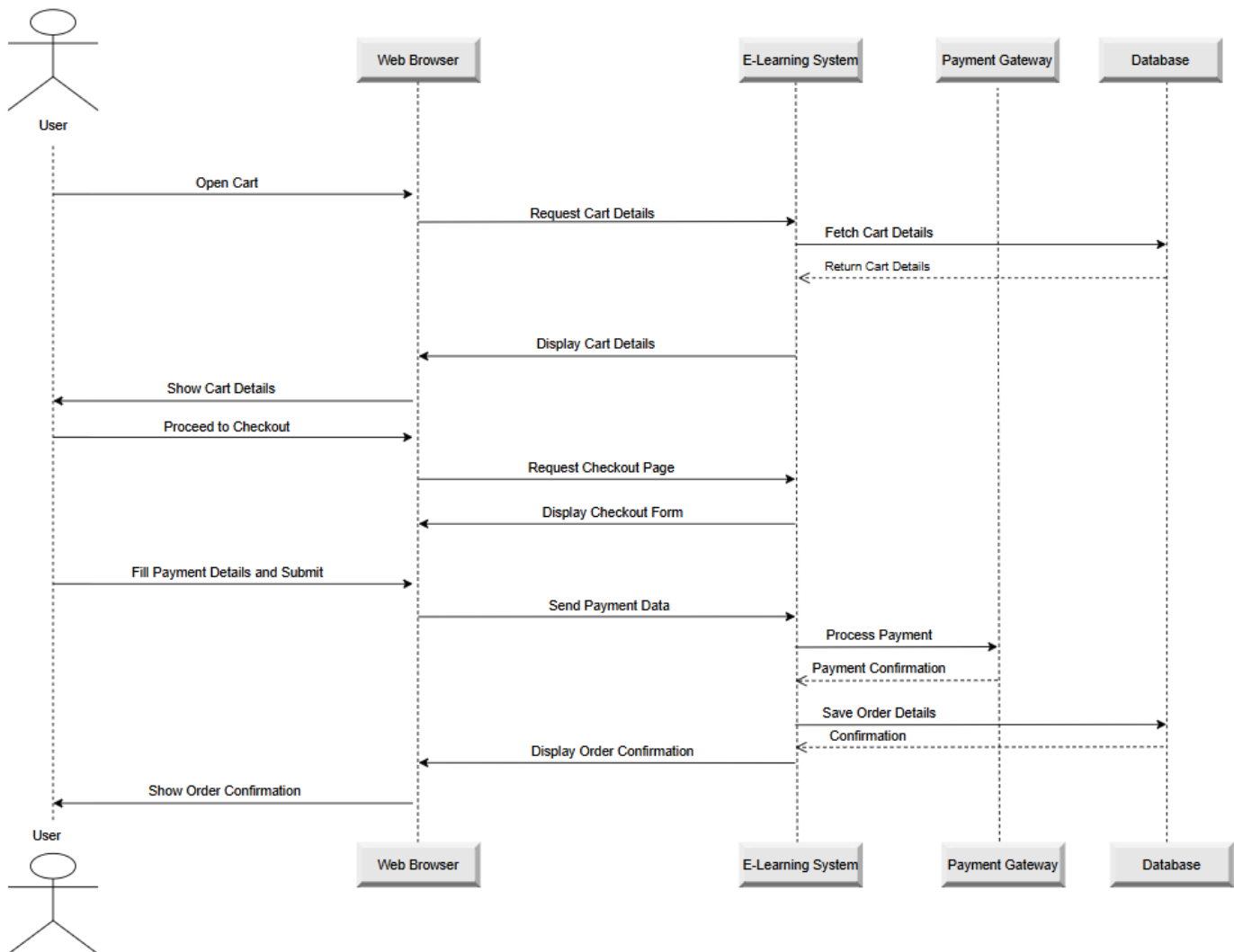


Figure 8: Sequence Diagram for Course and Payment

2.4.4 Sequence Diagram for Course Purchasing and Adding to Cart

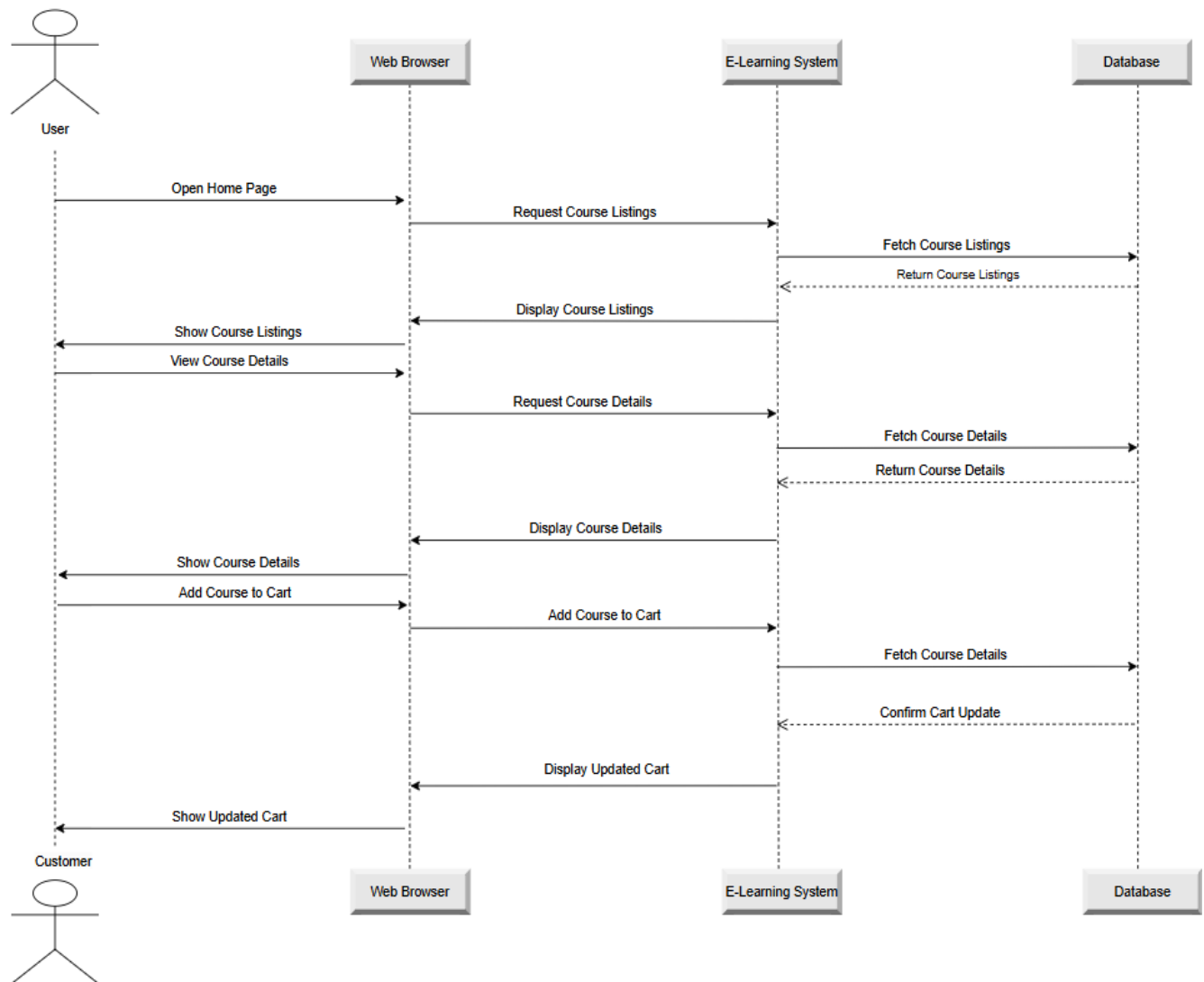


Figure 9: Sequence Diagram for Course Purchasing and Adding to Cart

2.5 Component Diagram

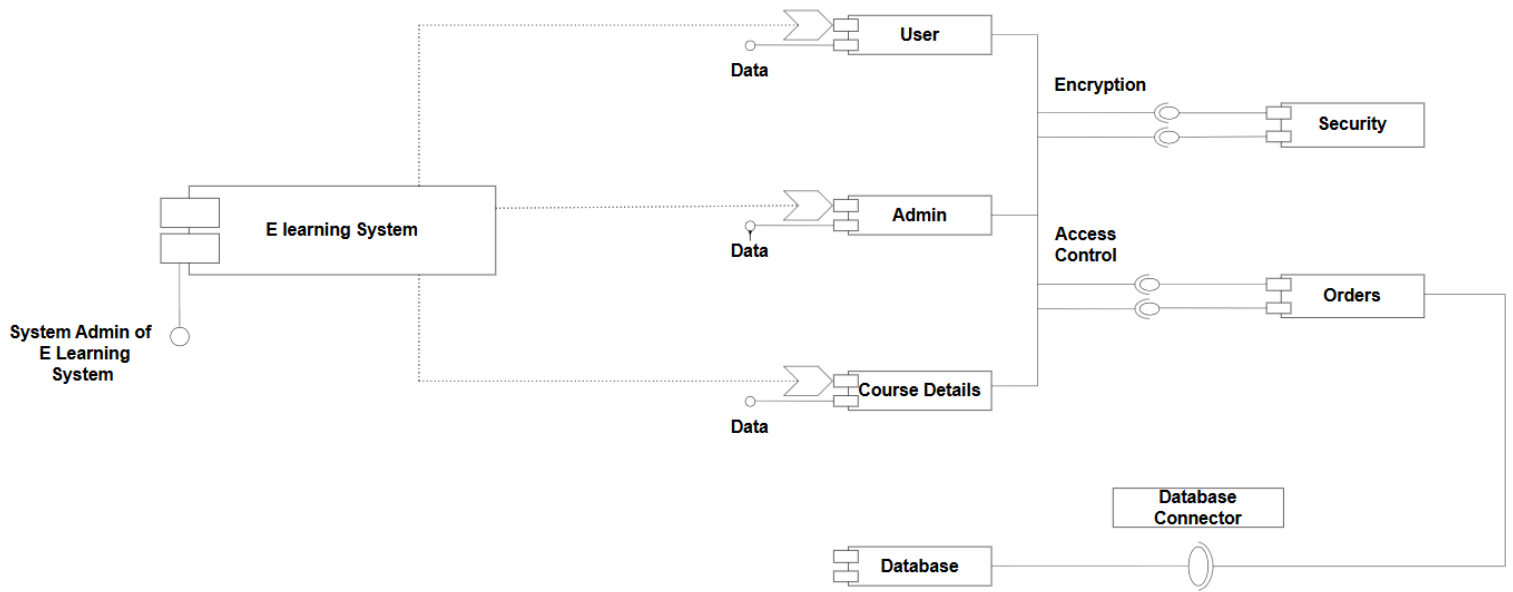


Figure 10: Component Diagram

2.6 Entity Relationship Diagram

An Entity Relationship (ER) Diagram, also known as an entity-relationship model, is a graphical representation of the relationships between different entities within a system. In the context of the proposed e-learning platform, an ER diagram illustrates the logical structure of the database by defining key entities such as courses, users, administrators, exams, reviews, lessons, certifications, and their respective attributes and relationships. This diagram provides a clear visual overview of how data is stored, organized, and connected within the system, ensuring efficient data management and retrieval.

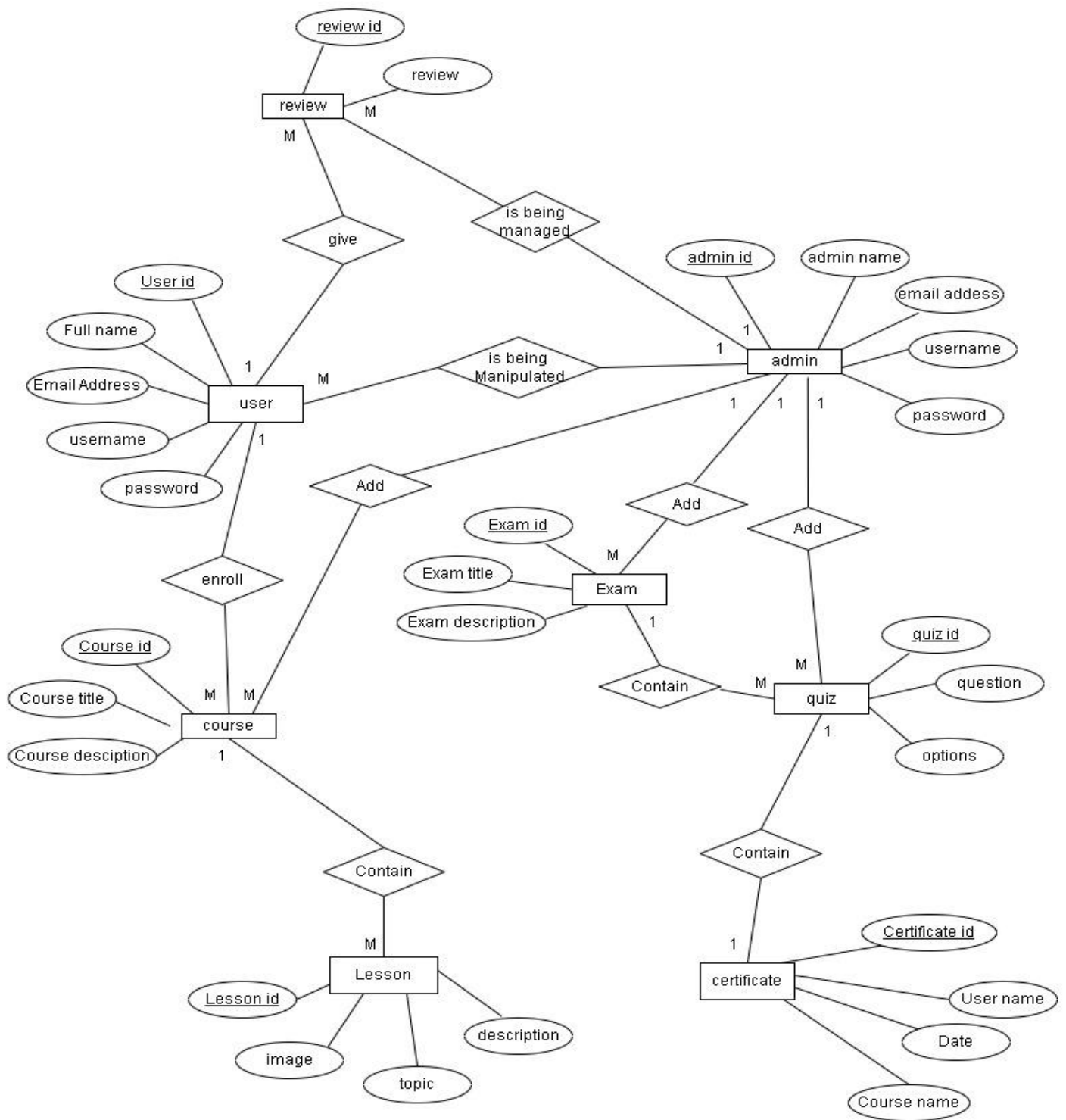


Figure 11: Entity Relationship Diagram

2.6.1 Relational Schema

Admin Table

Admin				
admin_id	admin name	email address	username	password

User Table

User					
user_id	full name	username	email address	phone	password

Course Table

Course		
course_id	course title	course description

Lesson Table

Lesson				
lesson_id	image	topic	description	course_id

Exam Table

Exam			
exam_id	exam_title	exam_description	exam_description

Quiz Table

Quiz			
quiz_id	question	options	exam_id

Certificate Table

Certificate					
certificate_id	user_name	course_name	date	user_id	course_id

Review Table

Review		
review_id	review	user_id

2.7 Summary

In summary of the above chapter, different types of UML diagrams are discussed, including use case diagrams, activity diagrams, sequence diagrams, and component diagrams. Use case diagrams are designed to capture the major requirements of the system, with use case narratives provided for each diagram. Activity and sequence diagrams are created to represent three key processes within the system. Additionally, entity relationship diagrams and relational schemas are designed for the web-based e-learning platform to depict the database structure and relationships among key entities.